Database Systems(CSL3050)

Lab Assignment 7 - Report

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Introduction

In this assignment, I interact with a database through queries using Java Database Connectivity (JDBC) to perform SQL queries. The database schema is from Assignment 6. The tasks include implementing methods to perform various database operations like retrieving employee lists, project assignments, and updating tables.In this assignment, I used IntelliJ IDEA to develop a Java application that interacts with a MySQL database through Java Database Connectivity (JDBC).

1 getList: Employee Details

The `getList` method lists all employees ordered alphabetically by their names. This method fetches employee data from the database and organizes it accordingly.

- Task: Retrieve the details of all employees sorted by their name.
- Query:

```
SELECT * FROM employee ORDER BY emp_name;
```

Screenshot:

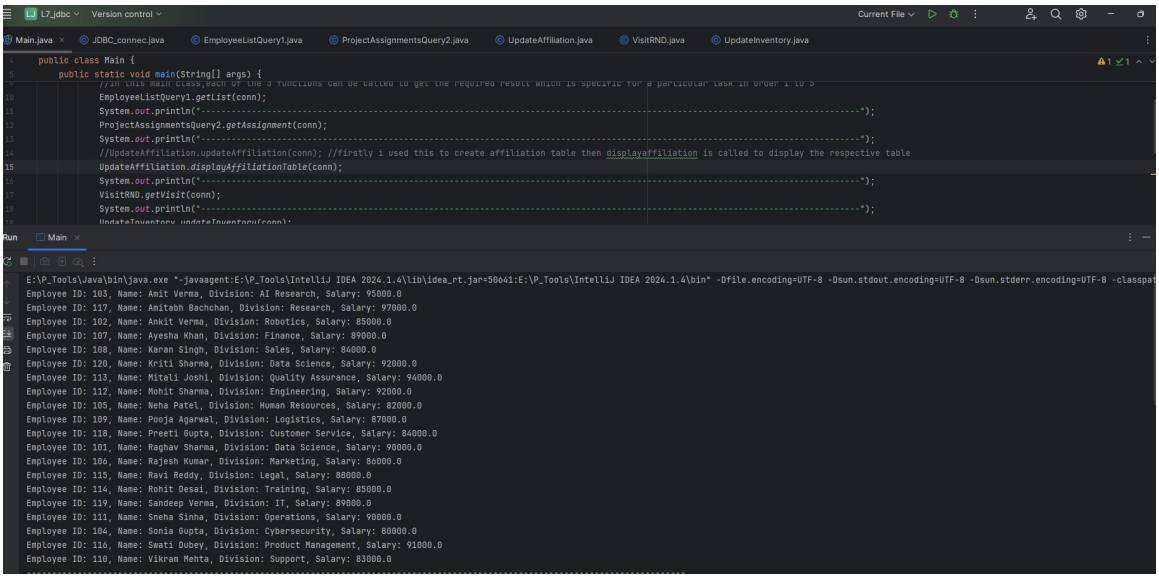


Figure 1: Employee List ordered by name

2 getAssignment: Project Assignments

The `getAssignment` method lists the details of all project assignments, including division and workspace information.

- Task: Retrieve project assignment details along with division and workspace.
- Query:

```
SELECT p.proj_id, p.proj_title, d.division_name, w.building_name
FROM project p
JOIN division d ON p.division_name = d.division_name
JOIN workspace w ON d.building_name = w.building_name;
```

Screenshot:

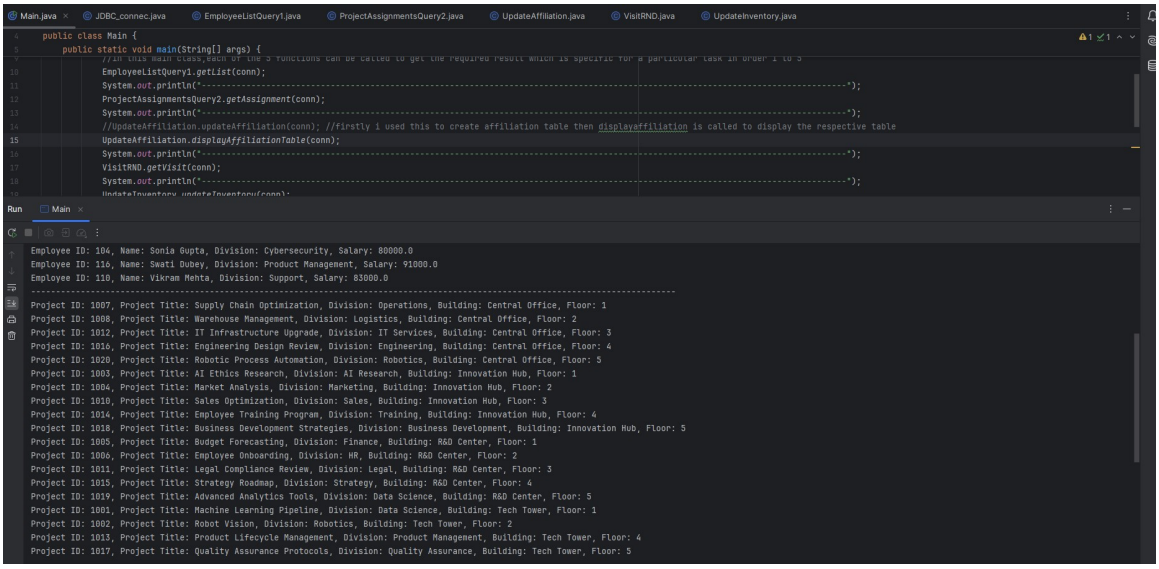


Figure 2: Project Assignments with Division and Workspace details

3 updateAffiliation: Employee Division Affiliation

The `updateAffiliation` method creates an `Affiliation` table to assign each employee to two different divisions. Changes are reflected in other related tables.

- Task: Create the `Affiliation` table and update records to associate each employee with two divisions.
- Query:

```
CREATE TABLE IF NOT EXISTS Affiliation (
  ID INT,
  division_id_1 VARCHAR(255),
  division_id_2 VARCHAR(255),
  PRIMARY KEY (ID),
  FOREIGN KEY (ID) REFERENCES employee(ID),
  FOREIGN KEY (division_id_1) REFERENCES division(division_name),
  FOREIGN KEY (division_id_2) REFERENCES division(division_name)
);
```

Screenshot:

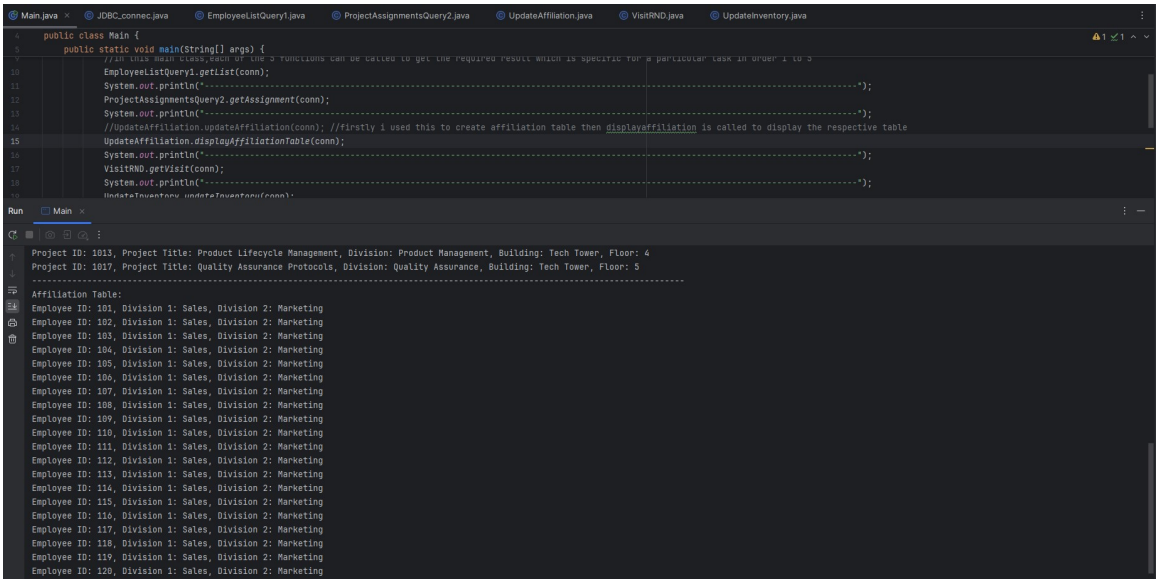


Figure 3: Affiliation Table Update

4 getVisit: R&D Division Project Visits

The `getVisit` method finds all employees and project details that were assigned to the R&D division between January and June 2022.

- Task: Retrieve all employees and project details for projects assigned to R&D from January to June 2022.
- Query:

```
SELECT e.emp_name, p.proj_title
FROM employee e
JOIN works_on w ON e.ID = w.ID
JOIN project p ON w.proj_id = p.proj_id
WHERE p.division_name = 'R&D'
AND (w.year = 2022 AND w.season BETWEEN 'January' AND 'June');
```

Screenshot:

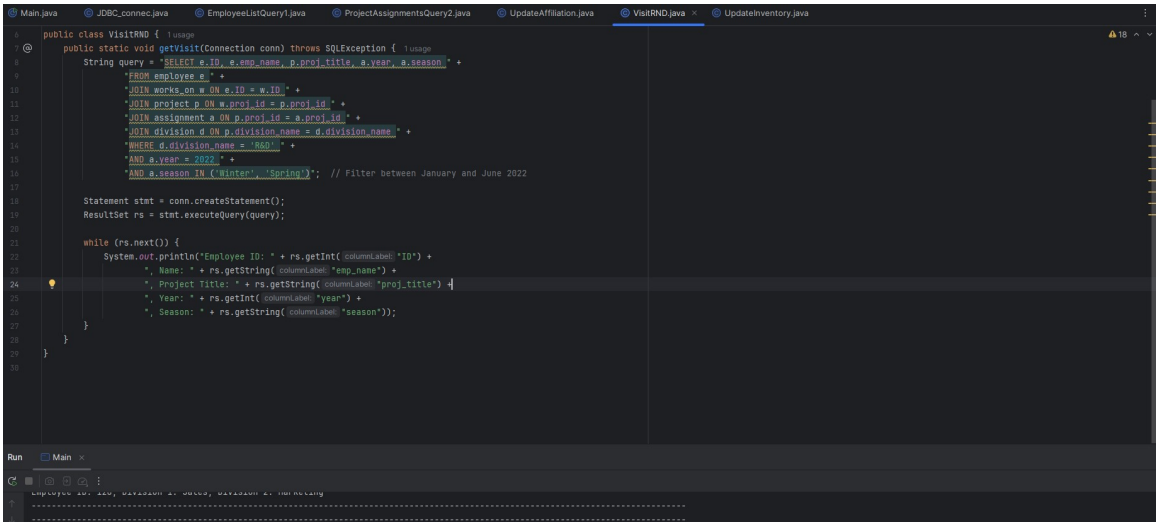


Figure 4: Employees assigned to R&D division projects between January and June 2022

5 updateInventory: Extend Project Duration and Increase Division Budget

The `updateInventory` method adds a `duration_months` column to the `project` table. It also increments the `operating_budget` of each division by 5% and extends project durations by 6 months.

- Task: Add a new column `duration_months`, update budgets and extend project durations.
- Query:

```
ALTER TABLE project ADD COLUMN IF NOT EXISTS duration_months INT;
UPDATE division
SET operating_budget = operating_budget * 1.05;
UPDATE project
SET duration_months = duration_months + 6;
```

Screenshot:

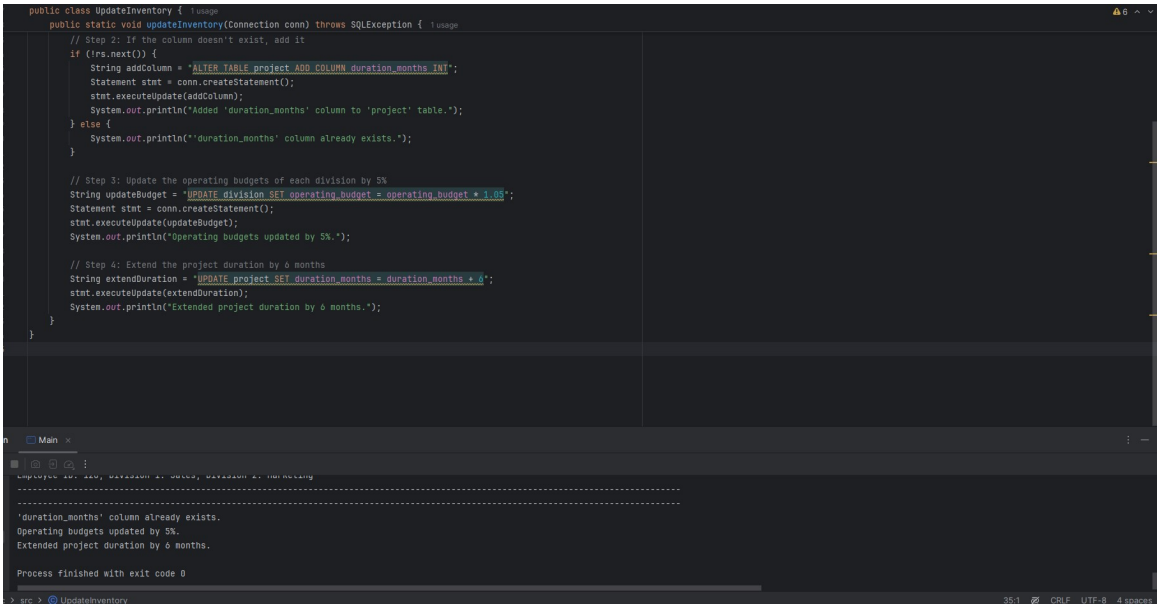


Figure 5: Inventory and Budget Update

Conclusion

This assignment demonstrates how to interact with a relational database using JDBC and perform a variety of SQL queries. The methods implemented handle tasks such as retrieving employee details, managing project assignments, updating affiliations, and modifying table structures like project durations and division budgets.